

DHIRAJ THAPA

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SUMMARY

Backend and full-stack engineer with hands-on experience building multi-tenant platforms, mobile applications, and distributed systems. Interested in large-scale infrastructure, API design, and engineering challenges at scale.

EDUCATION

Bachelor of Engineering in Computer Engineering **2021 – Present**
Institute of Engineering, Pulchowk Campus, Tribhuvan University | Aggregate: 78%

Higher Secondary Education (+2 Science) **2019 – 2021**
Trinity International Secondary School | GPA: 3.78/4.0

PROFESSIONAL EXPERIENCE

Mobile Application Developer | *AiGeeks*, UAE **Jan 2026 – Present**

- Building a multi-tenant analytics and survey platform targeting Gen Z audiences, serving brands, universities, and research teams
- Developing the React Native companion app as the primary participant-facing surface, handling survey delivery, rewards, and consent flows
- Contributing to Next.js frontend and Express + TypeScript backend, including RBAC, JWT authentication, and notification systems
- Designing PostgreSQL schemas for multi-tenant data isolation and cohort-based analytics queries

Full Stack Software Developer | *BrihaTech*, Kathmandu **Nov 2025 – Dec 2025**

- Built Digital Maitighar, a civic engagement platform enabling virtual protests and verified protest archives
- Implemented OAuth2.0 authentication and PostgreSQL for secure user management; built RESTful APIs for real-time interaction

PROJECTS

StoryGen – Full-Stack AI Storybook Generator *Next.js, TypeScript, Python, ComfyUI, Flux, Claude, Lightning AI*

- Developed an end-to-end generative AI pipeline that transforms a single child's drawing into a consistent 10-page illustrated storybook
- Engineered a multi-stage Python orchestration pipeline integrating SDXL (character enhancement), Claude Haiku (structured story generation), and Flux Kontext (scene illustration)
- Built a responsive Next.js web application with a Bun-based backend, featuring an asynchronous job-polling system to manage 10-minute long-running GPU workflows
- Achieved visual character consistency across multiple scenes without per-user fine-tuning by leveraging Flux Kontext with ControlNet/IP-Adapter conditioning
- Implemented a sequential LLM chain using Pydantic schema validation for narrative coherence and visual-brief readiness for downstream image models

Navajeevan – Second-hand Marketplace

- Full-stack marketplace with seller chat, payment integration, and dynamic inventory management
- OAuth2.0 authentication and MongoDB for secure user management and data persistence
- Built with Next.js and RESTful APIs for real-time item feeds, listing, and purchase handling

Blackground – Decentralized Social Media Platform

- Built on AT Protocol infrastructure with user data ownership and custom feed generation
- Architected backend with event-sourcing approach to preserve immutable content history on decentralized nodes

RetroFusion – Motion-Controlled Gaming Controller *Runner-up, Old Mobile Phone Reuse Competition (DoECE 2024)*

- Created an immersive gaming system using smartphone sensors to control 3D game environments in real-time
- Developed with Godot Engine for gameplay and .NET MAUI for cross-platform sensor communication

Urban Labyrinth – Pathfinding Simulation

- Designed a 3D virtual map of Pulchowk Campus implementing A* pathfinding for optimal navigation
- Built accurate 3D campus representation using Godot Engine with precise infrastructure simulation

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C++, Dart
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, GraphQL
Frontend & Mobile: Next.js, React Native (Expo)
Databases: PostgreSQL, MongoDB
AI / ML: PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, ComfyUI, Flux, ControlNet
Tools & Platforms: Git/GitHub, Docker, JWT/OAuth2.0, Traefik, RBAC systems
Other: System Design, UI/UX Design, Algorithm Development, Vivado/Vitis SDK

ACHIEVEMENTS

- **2nd Runner-up**, IEEE Computer Society Hack-a-flag CTF Competition (2024) – Analyzed security logs and identified vulnerabilities in capture-the-flag challenges
- **Runner-up**, Old Mobile Phone Reuse Competition, DoECE Pulchowk Campus (2024)

LEADERSHIP & EXTRACURRICULAR

Vice President – Pulchowk Music Club and Research Center **2024**
• Led organization of Pulchowk Music Fest 2024, coordinating logistics, sponsorships, and event execution for 200+ attendees
• Conducted Guitar Workshop mentoring 50+ students in music theory and performance techniques

Design Lead – SEDS Pulchowk **2023**
• Created visual content for events and campaigns; maintained consistent branding across club communications